Satellite Based Observation System for NSR



Team 3

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Mission Objectives



1. ICE AND ENVIRONMENTAL MONITORING
CONTINUOUSLY OBSERVE ICE COVERAGE, THICKNESS, AND ENVIRONMENTAL PARAMETERS ALONG THE NSR.



2. DATA TRANSMISSION
PROVIDE REAL-TIME DATA COMMUNICATION BETWEEN THE SATELLITE SYSTEM AND SHIPS NAVIGATING THE NSR.



3. RISK MITIGATION
USE ADVANCED SENSORS TO MINIMIZE ICE-RELATED RISKS AND SUPPORT SAFE NAVIGATION.

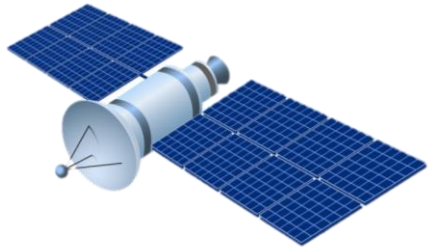


4. STAKEHOLDER DATA SHARING
FACILITATE COLLABORATION AND DATA SHARING BETWEEN GOVERNMENTS, SHIP OPERATORS, AND RESEARCHERS TO IMPROVE SAFETY AND SUSTAINABILITY.



5. ARCTIC-READY CUBESAT DESIGN
ENSURE THE SATELLITE SYSTEM IS RESILIENT TO ARCTIC WEATHER CONDITIONS, GUARANTEEING UNINTERRUPTED SERVICE.

Concept of Operation



4 Downlink

1 Downlink

Uplink 2





Ground Station receives the data (ice thickness, and position) and processes it.

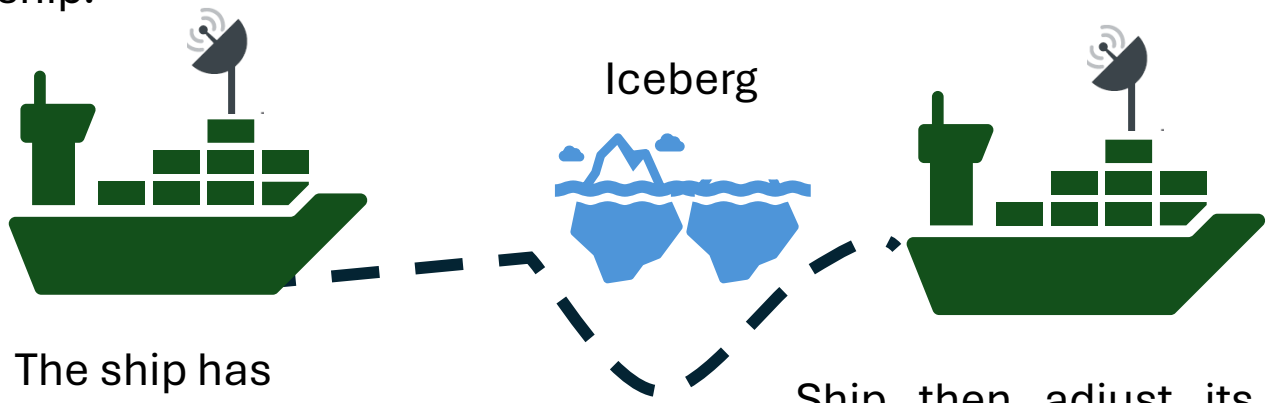
Ground Station sends the process data to headquarters and back to satellite.

3



Ship Companies (Sovcomflot, Rosatomflot)

The satellite then sends signal to the ship.

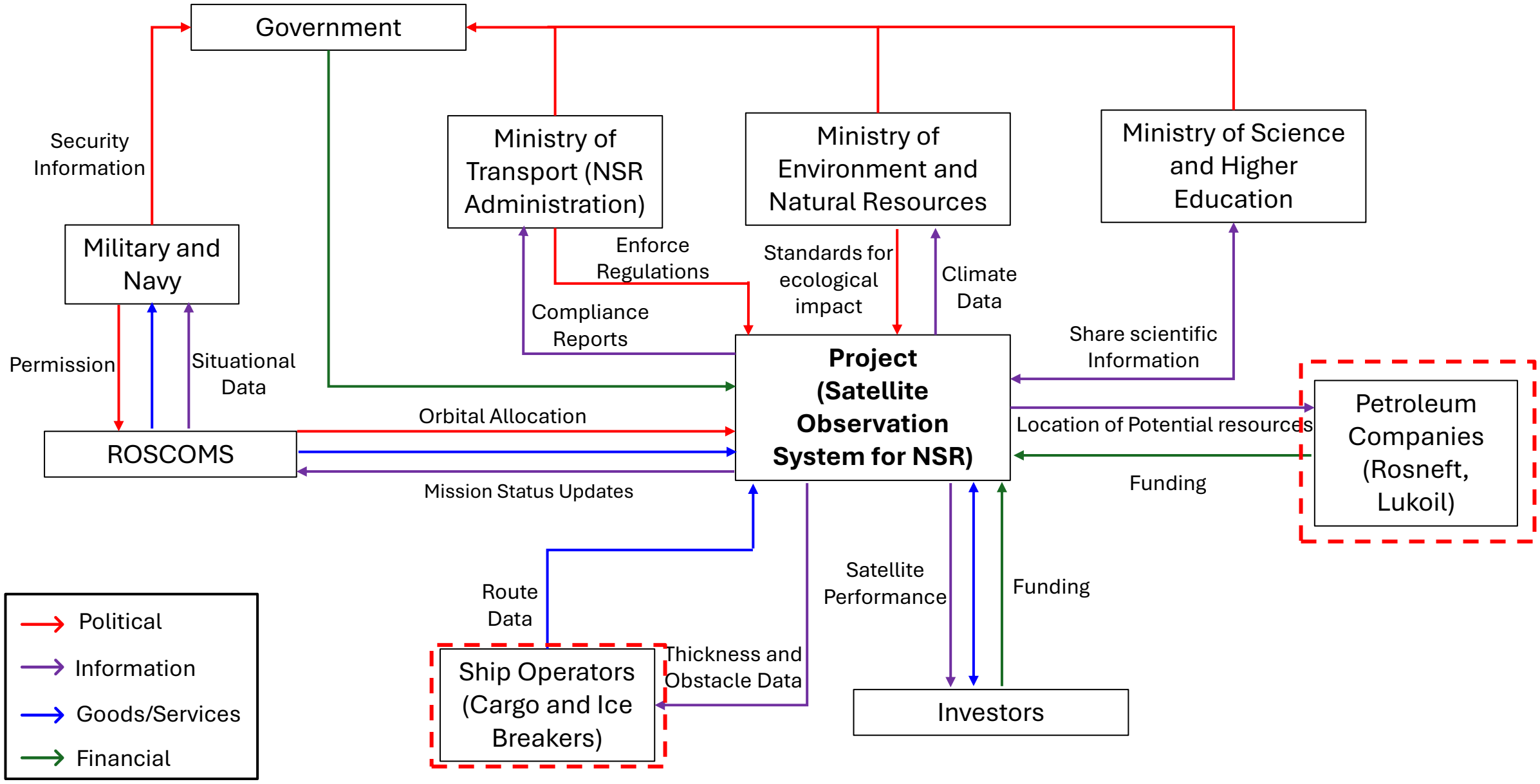


The ship has system for two-way communication.

Ship then adjust its route according to data.

5

Stake Holder Value Network



Stakeholder Interests and Goals Expectations

Government

-Establish infrastructure to ensure year-round navigation along NSR.

- Minimize risks of accidents and delays.

Military and Navy

Secure data transmission and real-time awareness for Arctic security needs.

Ministry of Transport (NSR Administration)

- Develop reliable navigation support systems for NSR traffic.
- Policies to improve transit time and cost-effectiveness.

Ministry of Environment and Natural Resources

-Monitoring of ice conditions to minimize ecological disturbances.

-Quality climate and environmental data to support Arctic studies and innovation

Ministry of Science and Higher Education

Open access to data for research purposes.

ROSCOMS

-Use space-based technologies to support NSR operations.

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Ship Operators (Cargo & Ice Breakers)

-Reduce operational costs and enable ship tracking and routing assistance.

Investors

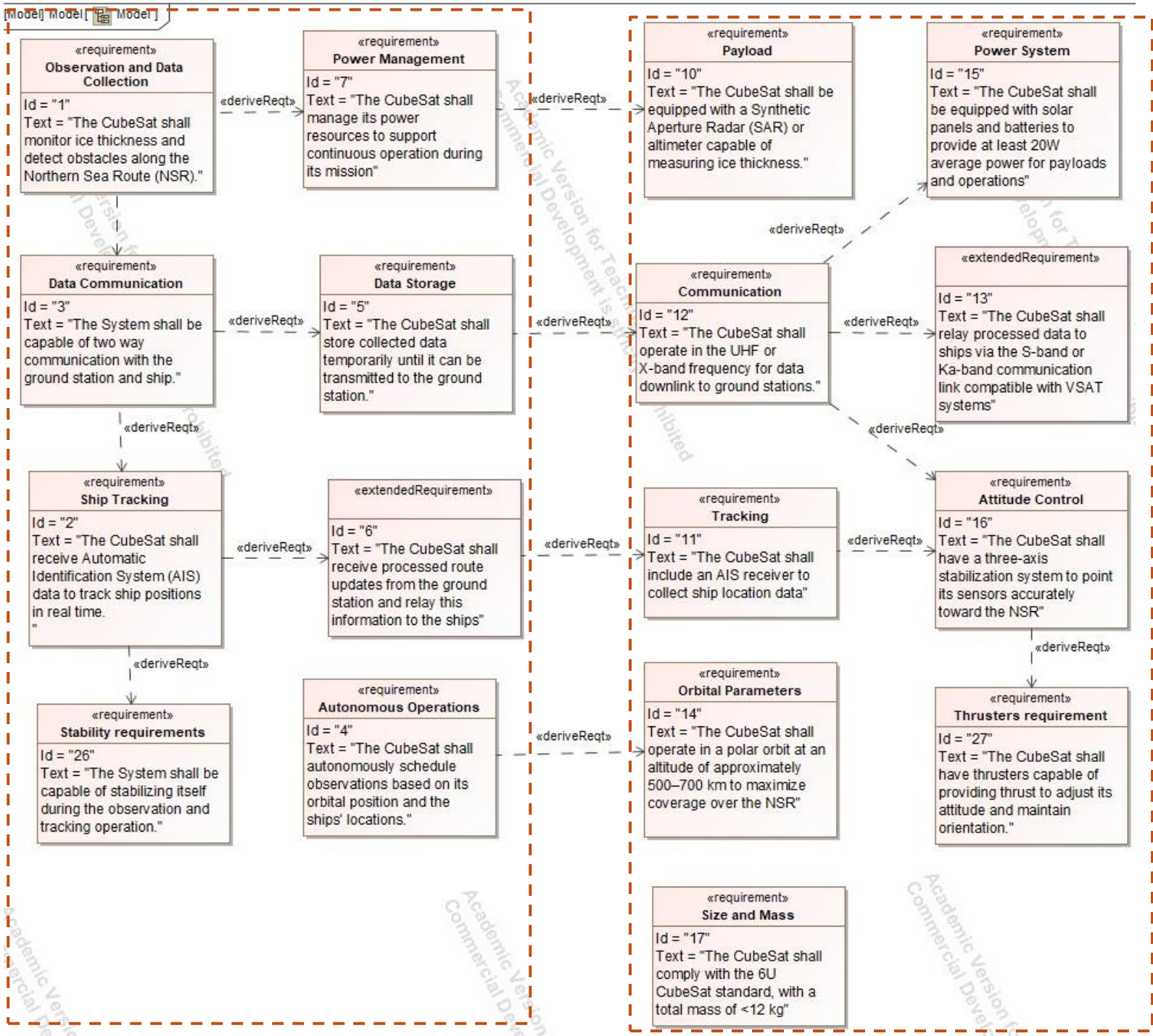
-Tangible **returns on investment**, budget transparency, and **project milestones**.

Petroleum Companies (Rosneft, Lukoil)

Resource mapping and develop safe routes for Arctic resource extraction.

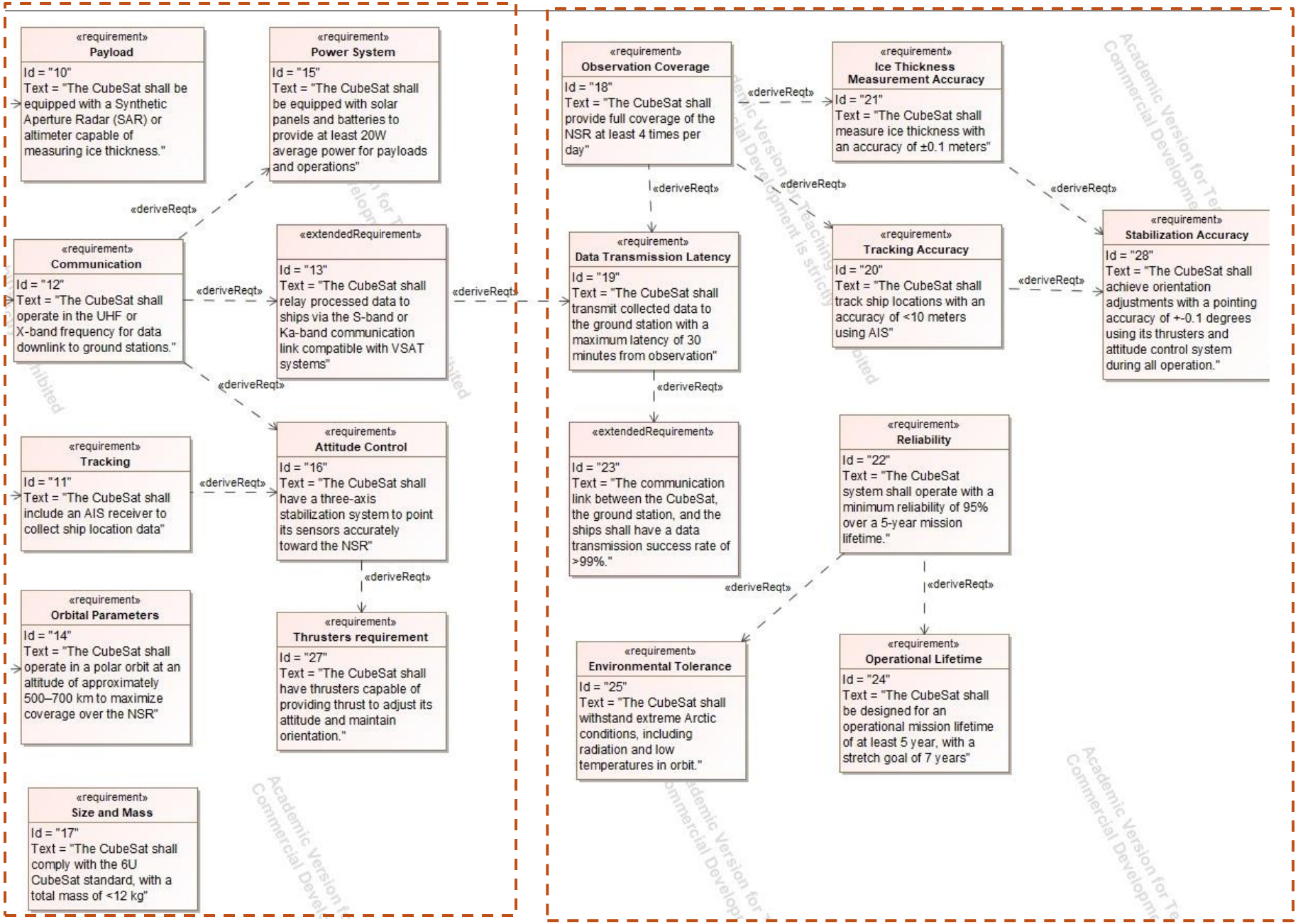
Requirements

Functional



Technical

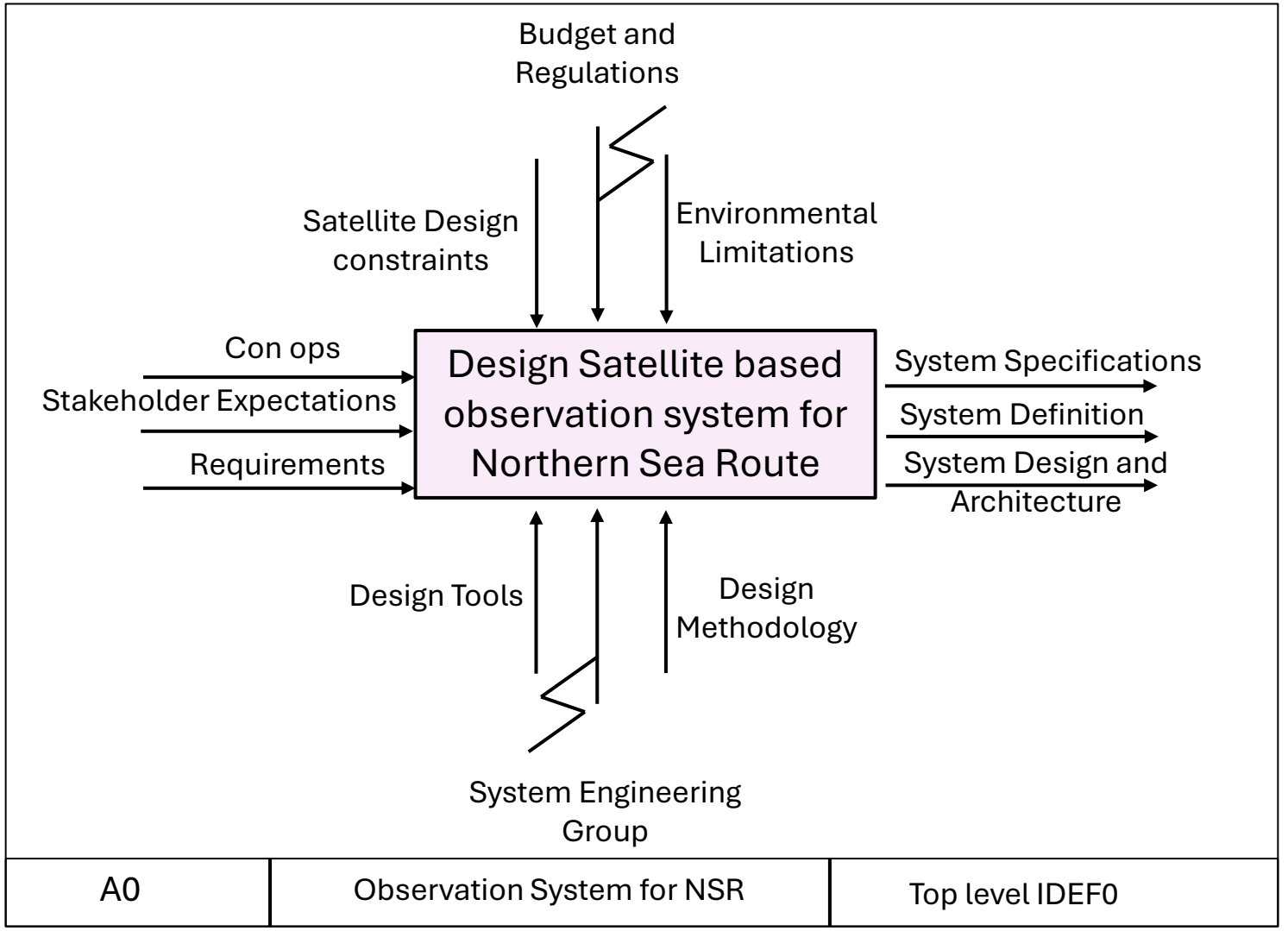
Continued...


Technical
Performance
«requirement»
Payload
Id = "10"
Text = "The CubeSat shall be equipped with a Synthetic Aperture Radar (SAR) or altimeter capable of measuring ice thickness."
«requirement»
Power System
Id = "15"
Text = "The CubeSat shall be equipped with solar panels and batteries to provide at least 20W average power for payloads and operations"
«deriveReqt»
«requirement»
Communication
Id = "12"
Text = "The CubeSat shall operate in the UHF or X-band frequency for data downlink to ground stations."
«deriveReqt»
«extendedRequirement»
Id = "13"
Text = "The CubeSat shall relay processed data to ships via the S-band or Ka-band communication link compatible with VSAT systems"
«deriveReqt»
«deriveReqt»
«requirement»
Tracking
Id = "11"
Text = "The CubeSat shall include an AIS receiver to collect ship location data"
«deriveReqt»
«requirement»
Attitude Control
Id = "16"
Text = "The CubeSat shall have a three-axis stabilization system to point its sensors accurately toward the NSR"
«deriveReqt»
«requirement»
Orbital Parameters
Id = "14"
Text = "The CubeSat shall operate in a polar orbit at an altitude of approximately 500–700 km to maximize coverage over the NSR"
«requirement»
Thrusters requirement
Id = "27"
Text = "The CubeSat shall have thrusters capable of providing thrust to adjust its attitude and maintain orientation."
«requirement»
Size and Mass
Id = "17"
Text = "The CubeSat shall comply with the 6U CubeSat standard, with a total mass of <12 kg"
«requirement»
Observation Coverage
Id = "18"
Text = "The CubeSat shall provide full coverage of the NSR at least 4 times per day"
«deriveReqt»
«requirement»
Ice Thickness Measurement Accuracy
Id = "21"
Text = "The CubeSat shall measure ice thickness with an accuracy of ±0.1 meters"
«deriveReqt»
«deriveReqt»
«deriveReqt»
«requirement»
Data Transmission Latency
Id = "19"
Text = "The CubeSat shall transmit collected data to the ground station with a maximum latency of 30 minutes from observation"
«requirement»
Tracking Accuracy
Id = "20"
Text = "The CubeSat shall track ship locations with an accuracy of <10 meters using AIS"
«deriveReqt»
«requirement»
Stabilization Accuracy
Id = "28"
Text = "The CubeSat shall achieve orientation adjustments with a pointing accuracy of +-0.1 degrees using its thrusters and attitude control system during all operation."
«deriveReqt»
«extendedRequirement»
Id = "23"
Text = "The communication link between the CubeSat, the ground station, and the ships shall have a data transmission success rate of >99%."
«requirement»
Reliability
Id = "22"
Text = "The CubeSat system shall operate with a minimum reliability of 95% over a 5-year mission lifetime."
«deriveReqt»
«deriveReqt»
«requirement»
Environmental Tolerance
Id = "25"
Text = "The CubeSat shall withstand extreme Arctic conditions, including radiation and low temperatures in orbit."
«requirement»
Operational Lifetime
Id = "24"
Text = "The CubeSat shall be designed for an operational mission lifetime of at least 5 year, with a stretch goal of 7 years"

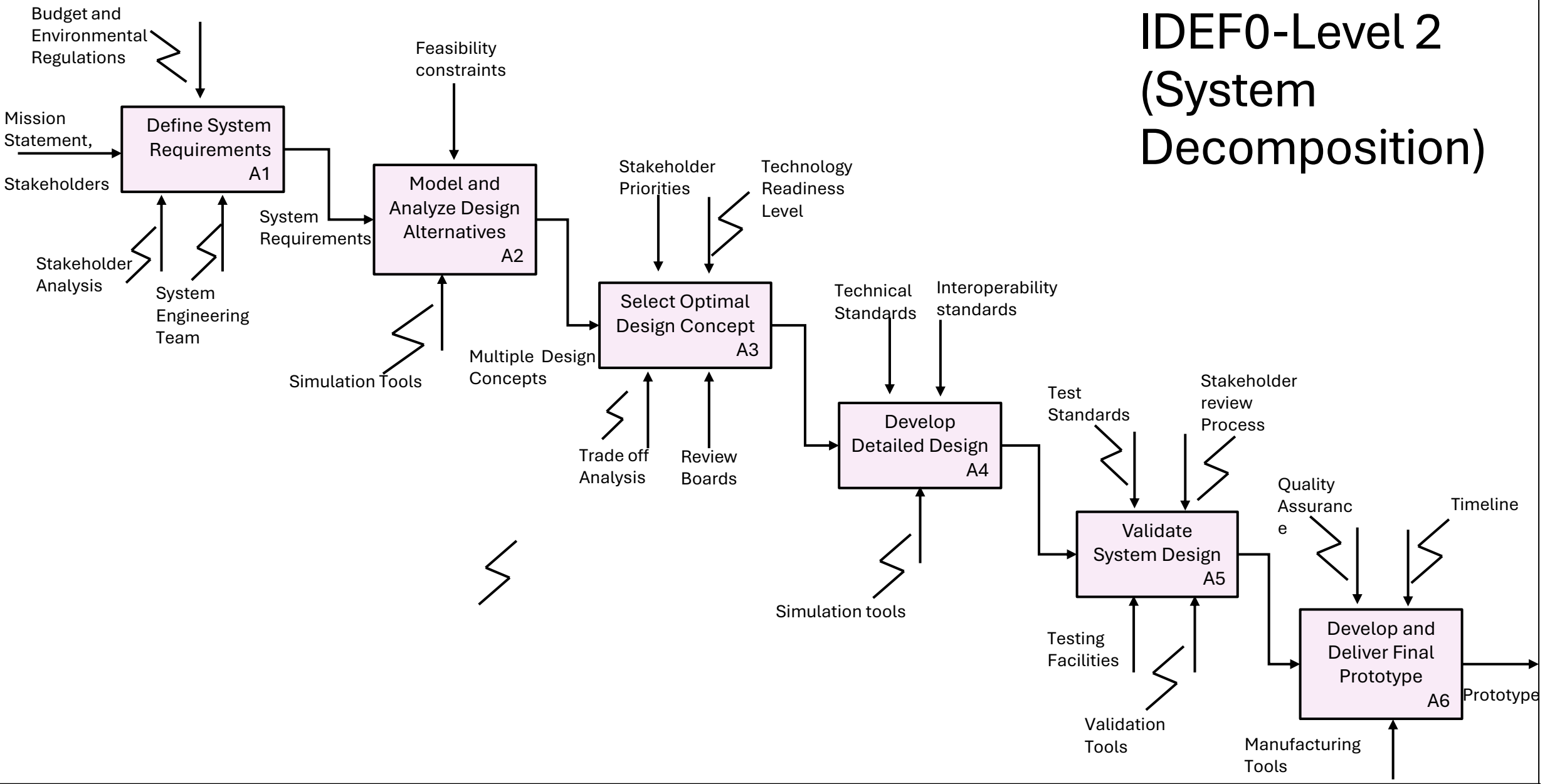
System Model IDEF0 (Top Level)

Purpose: To design and validate a CubeSat-based observation system for monitoring ice thickness along the Northern Sea Route to support safe and efficient ship navigation

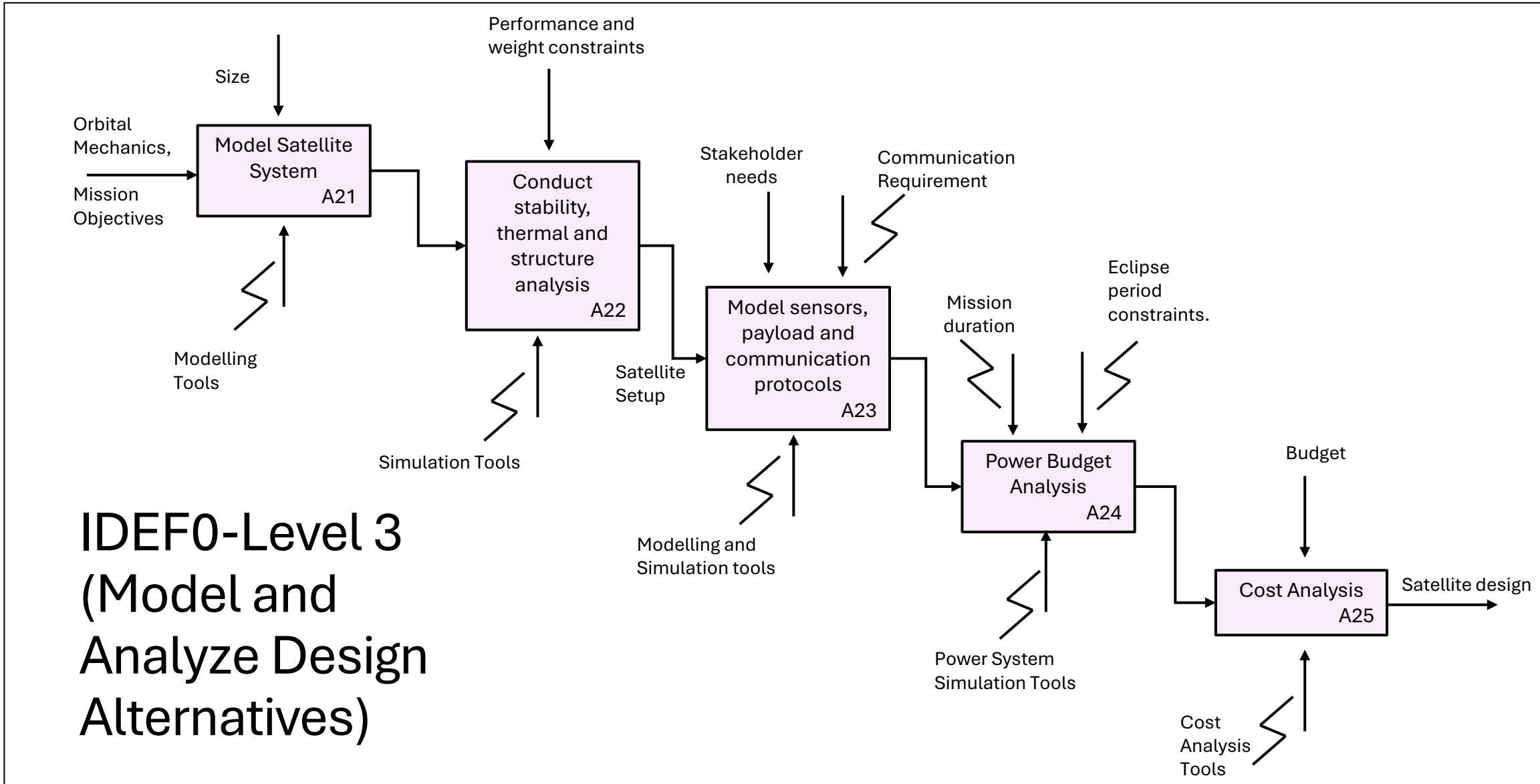
Point of View: Systems engineering perspective, focusing on stakeholder requirements, subsystem integration, and operational feasibility.



IDEF0-Level 2 (System Decomposition)



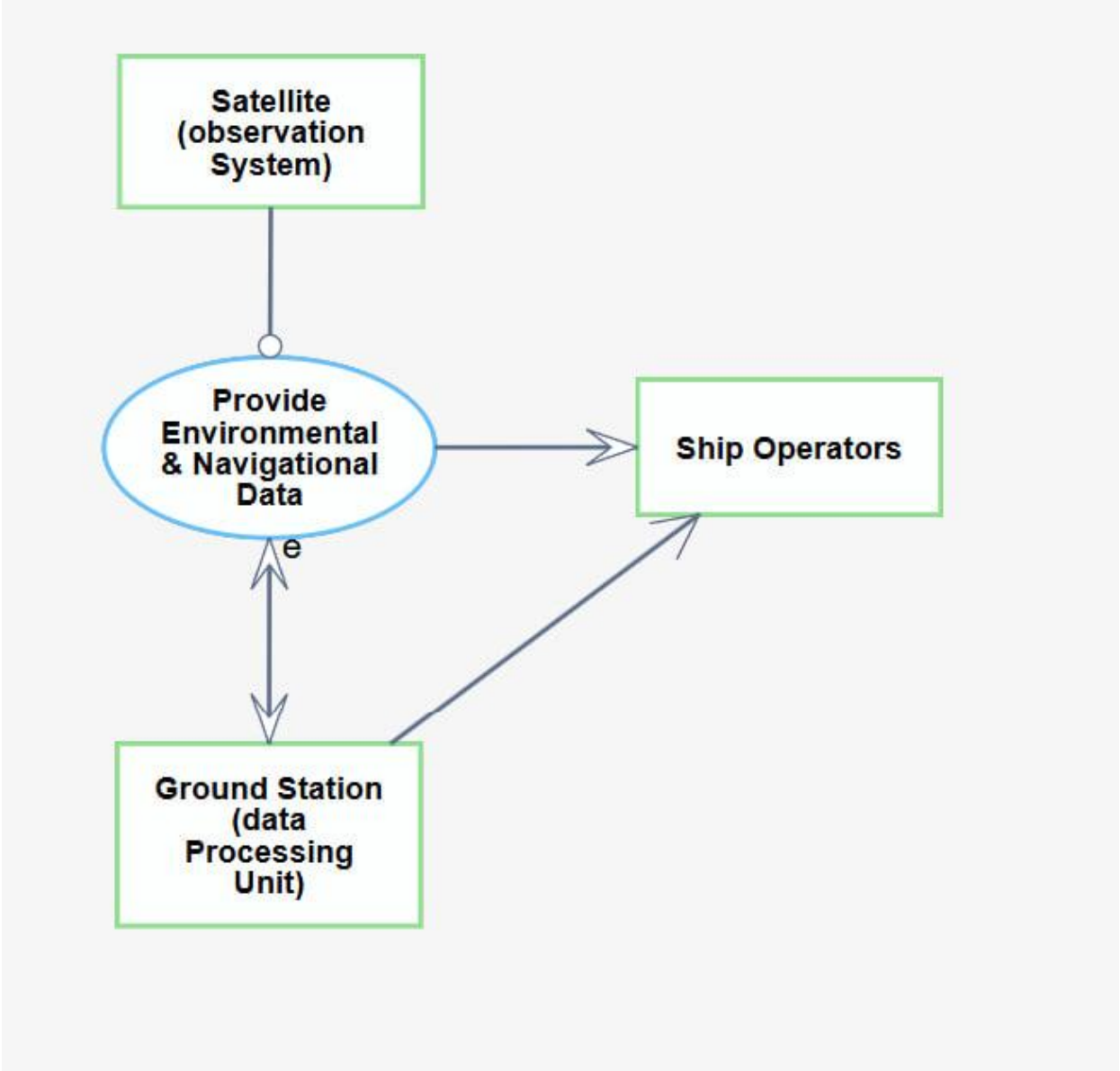
A0	Observation System for NSR	Level 2 IDEF0



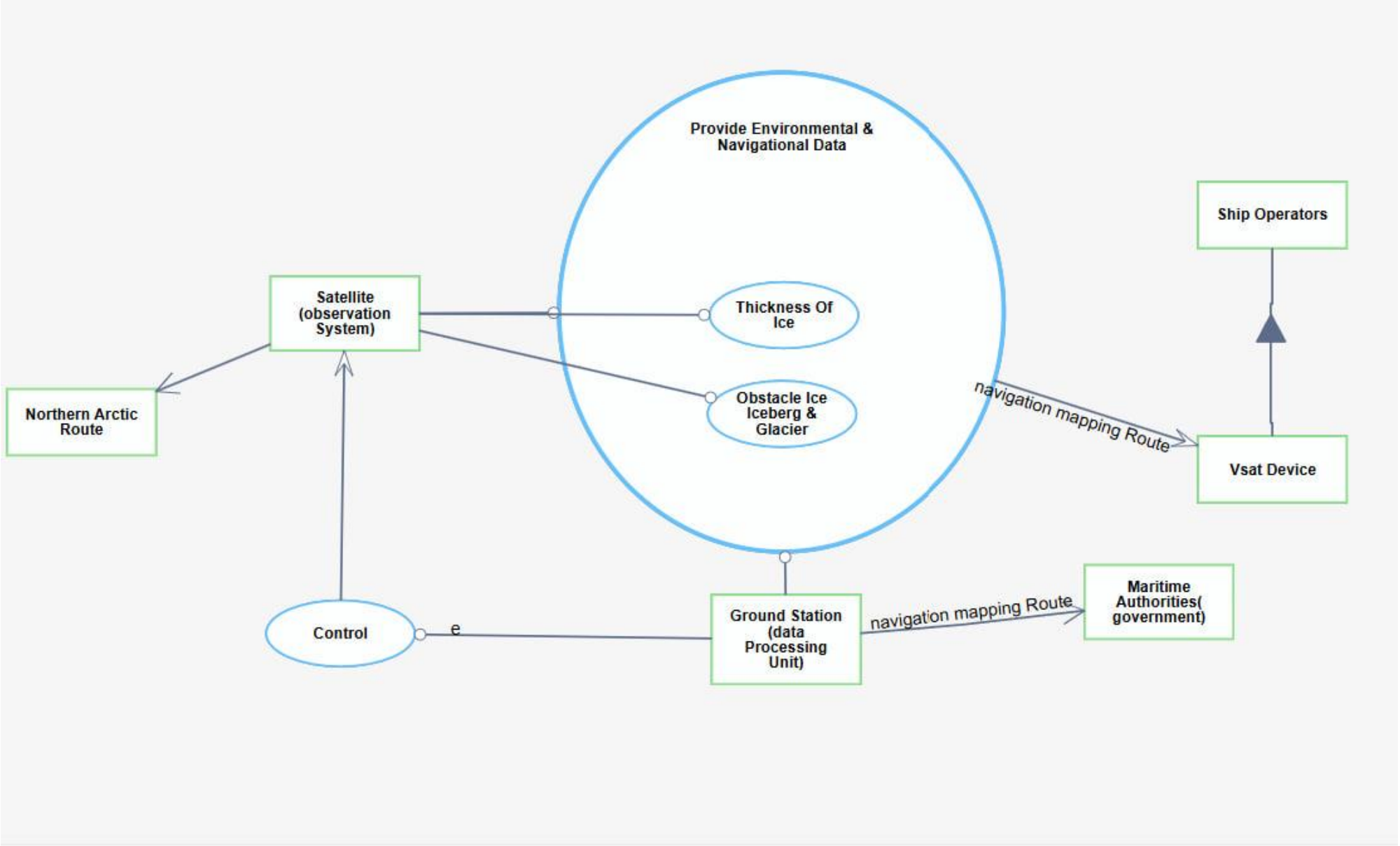
IDEF0-Level 3 (Model and Analyze Design Alternatives)

A2	Model and Analyze Design Alternatives	Level 3 IDEF0

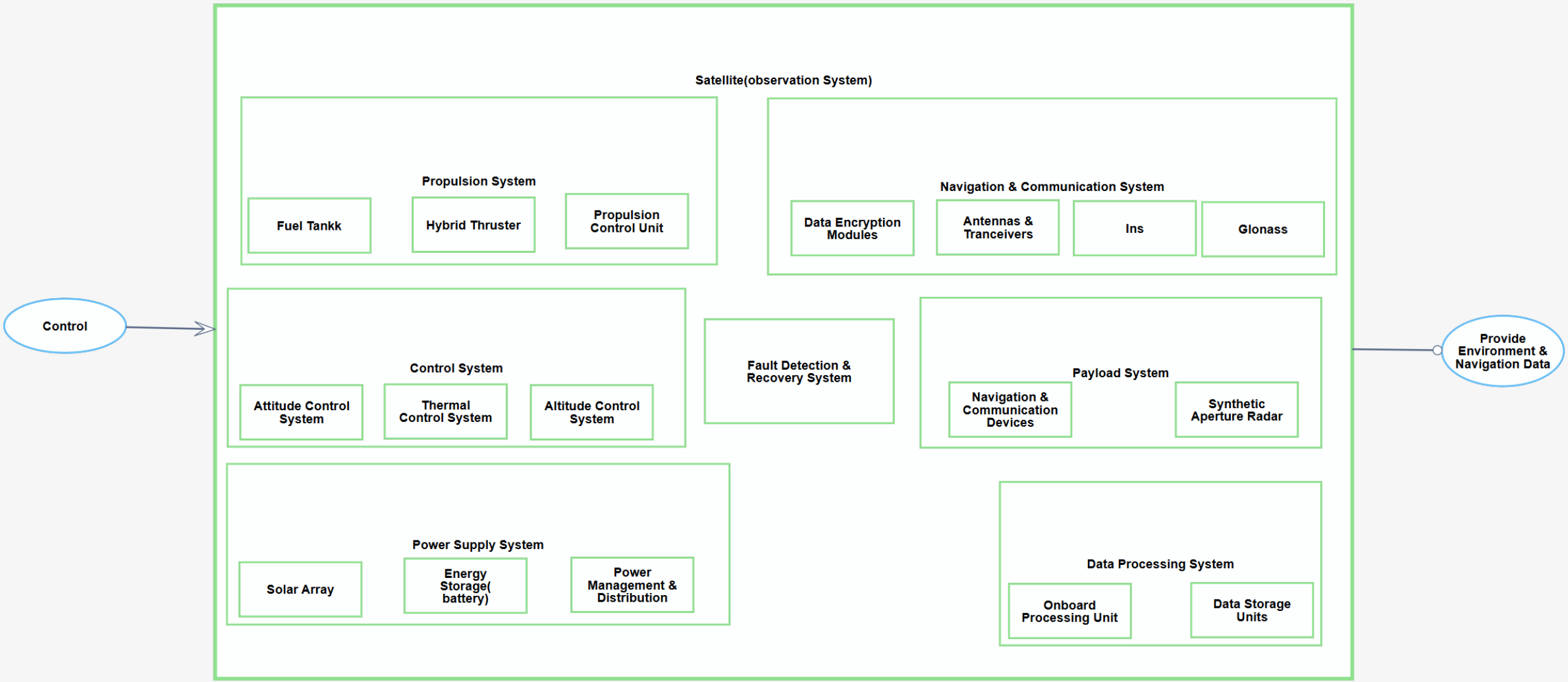
OPM Model


Satellite (observation System)
Provide Environmental & Navigational Data
e
Ship Operators
Ground Station (data Processing Unit)

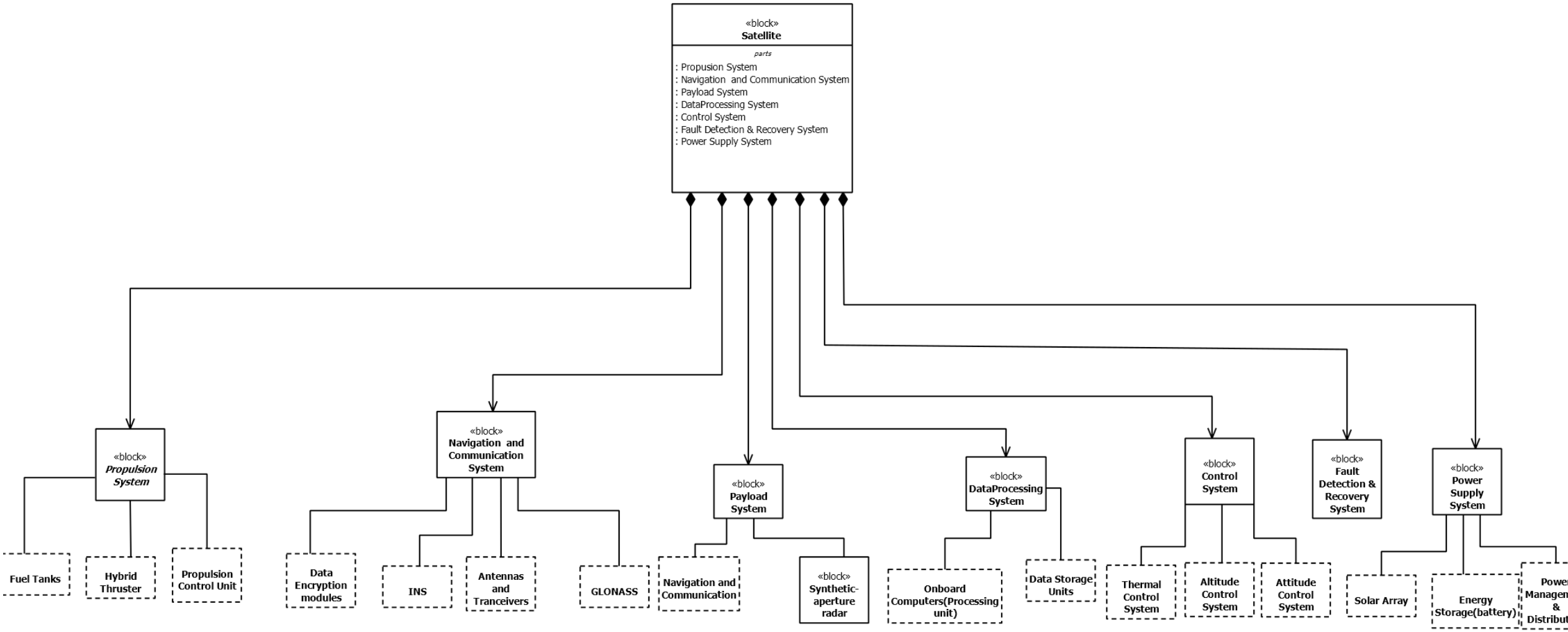
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Provide Environmental & Navigational Data
Ship Operators
Satellite (observation System)
Thickness Of Ice
Obstacle Ice Iceberg & Glacier
Northern Arctic Route
navigation mapping Route
Vsat Device
Control
e
Ground Station (data Processing Unit)
navigation mapping Route
Maritime Authorities(government)

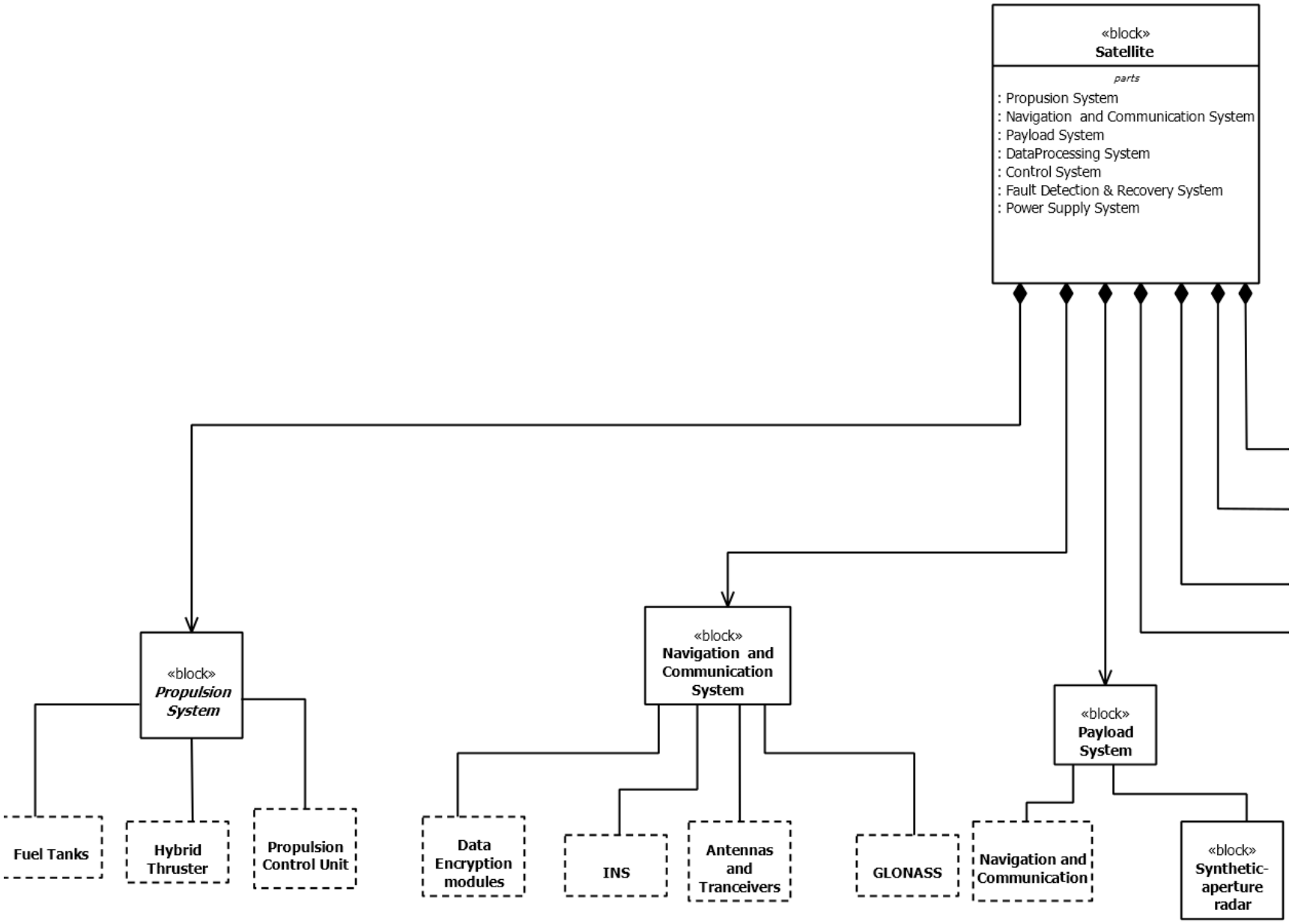
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Satellite(observation System)
Propulsion System
Fuel Tankk
Hybrid Thruster
Propulsion Control Unit
Navigation & Communication System
Data Encryption Modules
Antennas & Tranceivers
Ins
Glonass
Control
Control System
Attitude Control System
Thermal Control System
Altitude Control System
Fault Detection & Recovery System
Payload System
Navigation & Communication Devices
Synthetic Aperture Radar
Provide Environment & Navigation Data
Power Supply System
Solar Array
Energy Storage(battery)
Power Management & Distribution
Data Processing System
Onboard Processing Unit
Data Storage Units

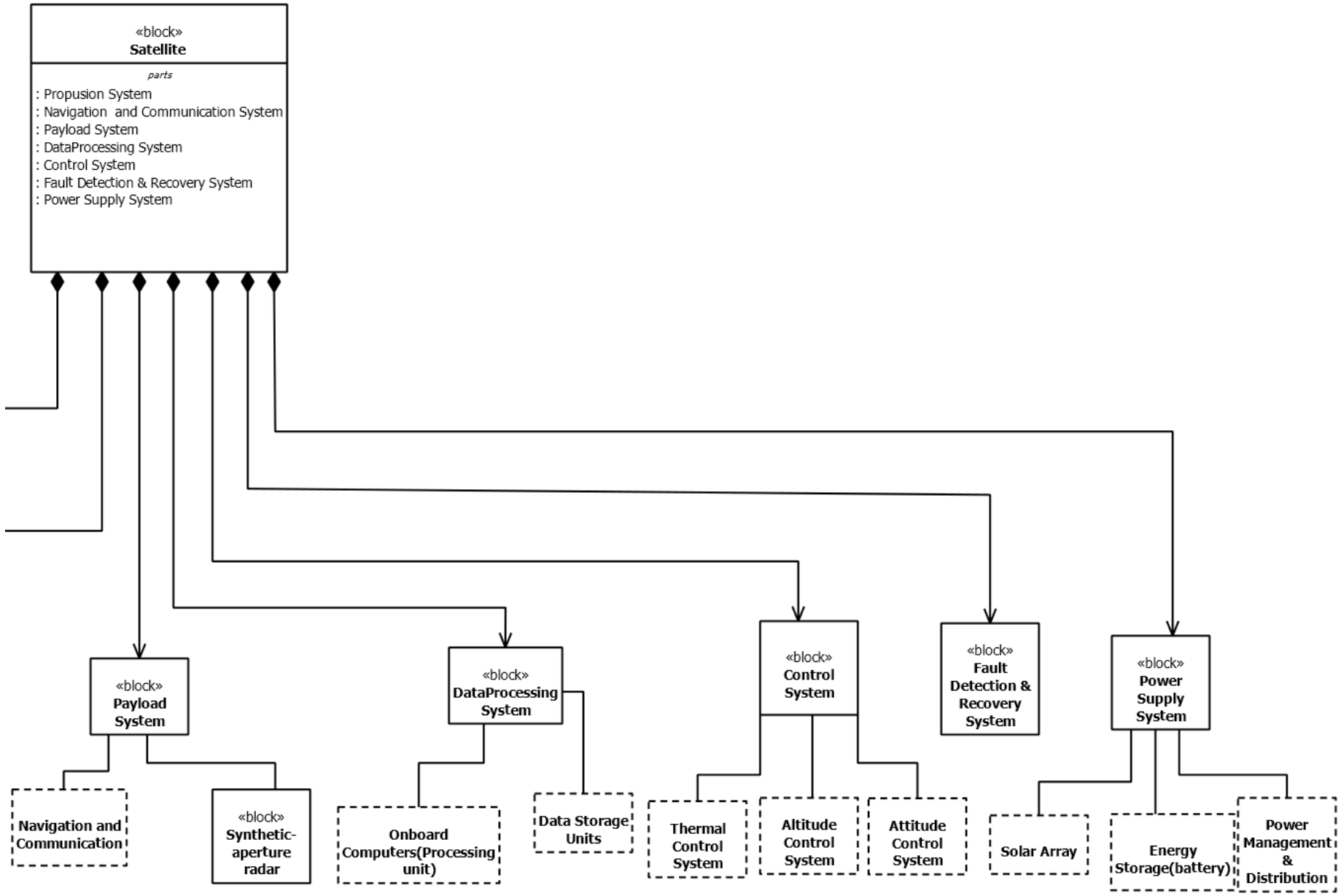
SysMl


«block»
Satellite
parts
: Propusion System
: Navigation and Communication System
: Payload System
: DataProcessing System
: Control System
: Fault Detection & Recovery System
: Power Supply System
«block»
Propulsion System
Fuel Tanks
Hybrid Thruster
Propulsion Control Unit
«block»
Navigation and Communication System
Data Encryption modules
INS
Antennas and Tranceivers
GLONASS
«block»
Payload System
Navigation and Communication
«block»
Synthetic-aperture radar
«block»
DataProcessing System
Onboard Computers(Processing unit)
Data Storage Units
«block»
Control System
Thermal Control System
Altitude Control System
Attitude Control System
«block»
Fault Detection & Recovery System
«block»
Power Supply System
Solar Array
Energy Storage(battery)

SysMl


«block»
Satellite
parts
: Propusion System
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: Control System
: Fault Detection & Recovery System
: Power Supply System
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Navigation and Communication System
Data Encryption modules
INS
Antennas and Tranceivers
GLONASS
«block»
Payload System
Navigation and Communication
«block»
Synthetic-aperture radar

SysMl


«block»
Satellite
parts
: Propusion System
: Navigation and Communication System
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: DataProcessing System
: Control System
: Fault Detection & Recovery System
: Power Supply System
«block»
Payload System
«block»
DataProcessing System
«block»
Control System
«block»
Fault Detection & Recovery System
«block»
Power Supply System
Navigation and Communication
«block»
Synthetic-aperture radar
Onboard Computers(Processing unit)
Data Storage Units
Thermal Control System
Altitude Control System
Attitude Control System
Solar Array
Energy Storage(battery)
Power Management & Distribution

Risk Analysis

- Financial Risks
- Technical Risks
- Industrial Risks
- Regulatory and Political Risks
- Environmental Risks

Technical Risks

Component Failure

Over time, devices such as solar panels and sensors can age or fail during operation

Communication Failures

Interference under different working conditions can cause delays in data transmission, or in more severe cases, complete loss of connection.

Software Bugs

The control system on CubeSat can become overloaded, leading to loss of communication.

Industrial Risks

Launching Failure

Rockets or spacecraft designated for low earth orbit satellites may fail to launch successfully, delaying the deployment of the satellite or, worse, completely destroying it.

Delayed Mission

Essential components for CubeSat may not be delivered on time by contractors, or the assembly of CubeSat may take longer than expected.

Regulatory and Political Risks

Compliance with Regulations

Non-compliance with regulations can lead to loss of credibility and difficulties in international agreements.

Political Changes

The aerospace policies of different countries can lead to disputes and affect the mission.

Environmental Risks

Weather Conditions

Under varying weather conditions, data collected from the satellite may not match the results obtained under good weather conditions. However, these discrepancies can also provide environmental information..

Meteorites and Space Debris

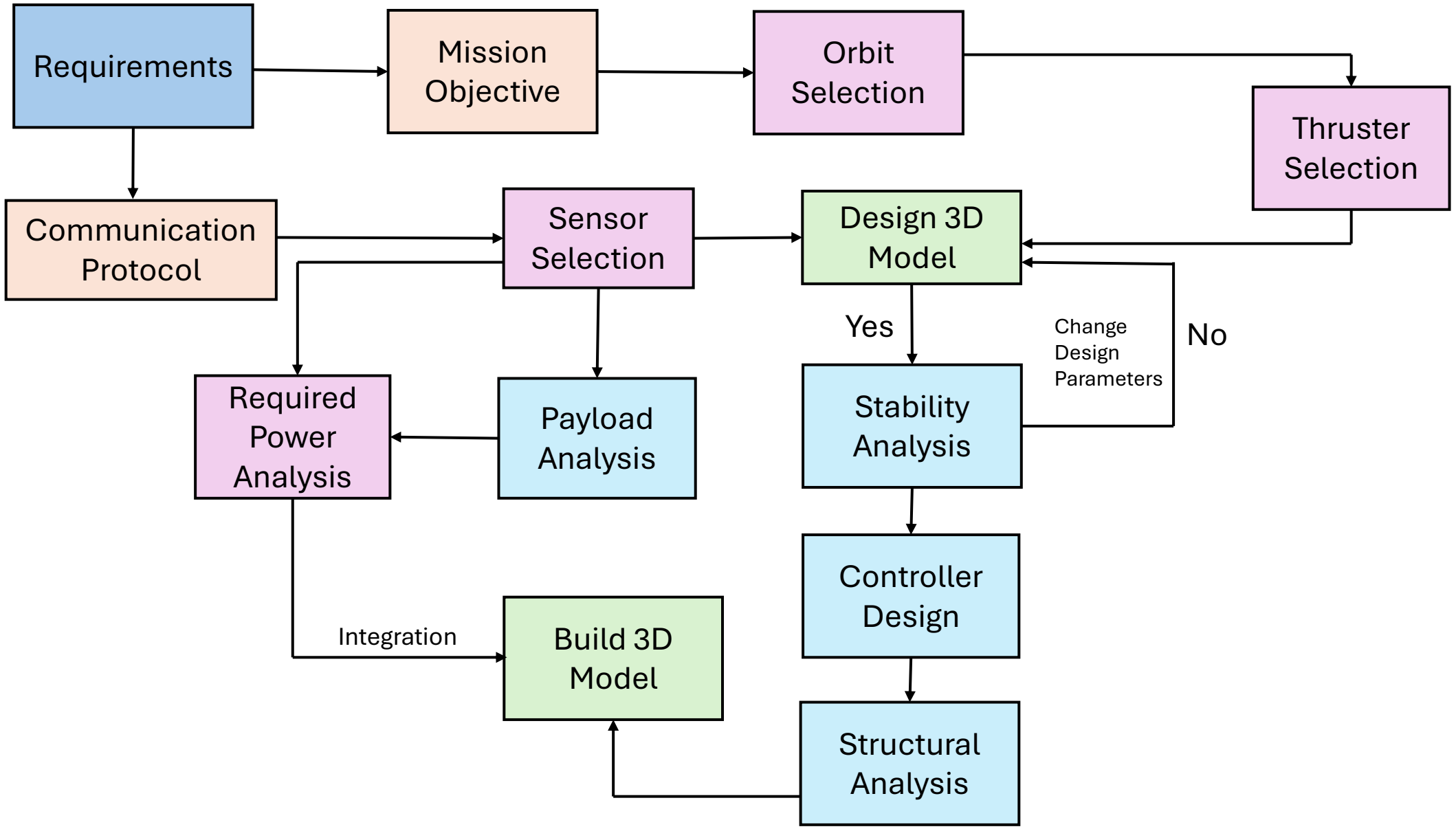
Collisions with meteorites or solar storms can cause partial or complete failure of the CubeSat system.

Risk Analysis

Risks			Prob.	Impact	Mitigation
Technical	Component Failure	T1	4	2	Testing under work conditions
	Communication Failures	T2	3	2	Improve telecommunications infrastructure and bandwidth
	Software Bugs	T3	2	2	Self-debug and Reboot System
Industrial	Launching Failed	I1	1	4	Having different contractors
	Mission Delayed	I2	2	4	Preparing spare parts
Regulatory and Political	Regulation Compliance	P1	2	3	Staying informed of satellite operational regulations and ensure compliance
	Political Changes	P2	2	3	Establish collaborative agreements for operations and data sharing
Financial	Financial Risks	F1	2	4	Cost managing and reschedule plan
Environmental	Weather conditions	E1	5	1	Using other CubeSat for data, collecting weather data
	Orbit Debris	E2	2	5	Using collision reporting to limit entry into high-risk collision areas
	Astronomical object	E3	1	5	Using space weather reports to avoid effects

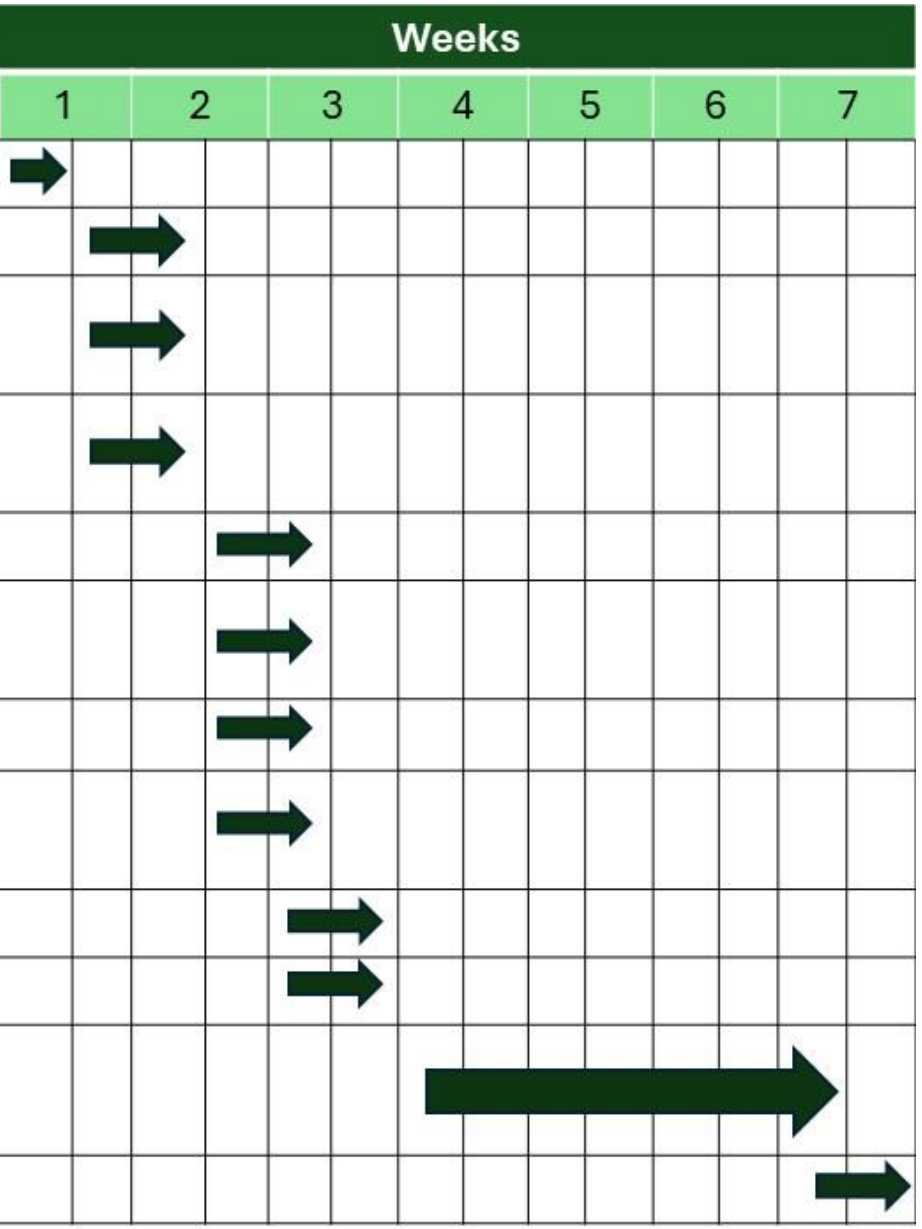
Prob. \ Impact	1	2	3	4	5
5	2(E1)	3	6	9	12
4	2	3(T1)	5	8	11
3	1	2(T2)	4	7	10
2	1	2	3(P1,P2)	5(I2,F1)	8(E2)
1	1	1	2	3(I1)	5(E3)

Prototyping Plan



Work Break Down Structure (Gantt Chart)

Task Name	Assigned To	Start Date	End Date	No. of Days
Mission Objectives	All	6-11-2024	7-11-2024	1
ConOps	Jeffrin	8-11-2024	12-11-2024	5
Stakeholder Analysis	Faryal	8-11-2024	12-11-2024	5
Stakeholders Expectations	Khang, Jeffrin	8-11-2024	12-11-2024	5
System Requirements	Duet	13-11-2024	18-11-2024	6
System Modelling	Faryal, Khang	13-11-2024	18-11-2024	6
System Architecture	Tayyaba	13-11-2024	18-11-2024	6
Risk Analysis and Mitigation	Alexander , Khang	13-11-2024	18-11-2024	6
Prototyping Plan	Tayyaba	19-11-2024	21-11-2024	3
PDR	All	22-11-2024	25-11-2024	4
Prototyping Hardware Software Validation	All	26-11-2024	13-12-2024	18
CDR	All	16-12-2024	20-12-2024	5



Team Structure

Team Leader	MBSE Modeler	Analyst	Configuration & Data Manager	Prototyping
Faryal Batool	Jeffrin Sam Faryal Batool	Khang Nguyen Saptadeep Ghosh	Faryal Batool Tayyaba Tanveer Alex	Saptadeep Ghosh Khang Nguyen


Thank You!